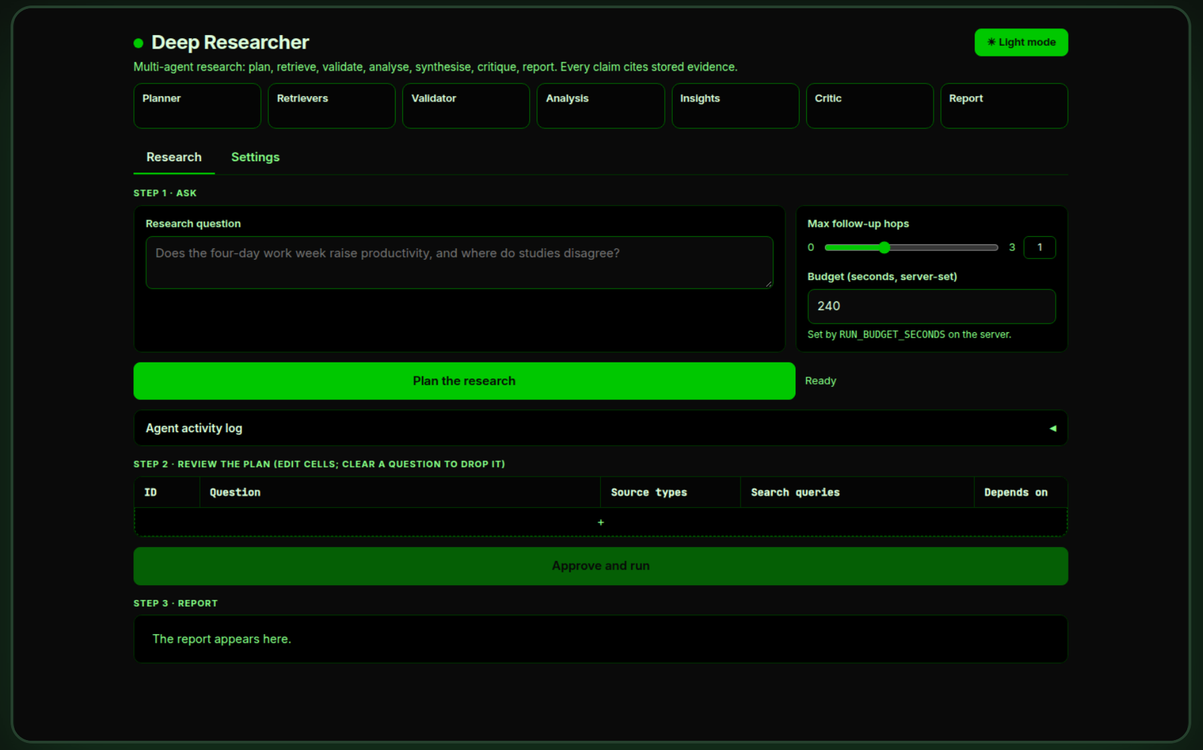
MULTI-AGENT RESEARCH ASSISTANT

Deep Researcher

A hard question in. A fully cited report out.

Seven agents plan, retrieve, validate, analyse, hypothesise, critique and write. Every claim cites stored evidence that is checked before the report ships.

LangGraph · OpenRouter · Chroma · FastAPI



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Seven agents, one cited report

01
Planner
Breaks the question into sub-questions

02 · human
You approve
Edit, drop or add plan rows

03
Retrievers ×N
Search papers, web and news in parallel

04
Validator
Grades relevance and credibility

↓ Chroma evidence store · 512-token chunks · 256-d vectors ↓

05
Analysis
Claims, contradictions, gaps

06
Insights
Hypotheses with reasoning chains

07
Critic
Red-team: accept, revise, dig deeper

08
Report
Every citation verified

LOOP · MULTI-HOP

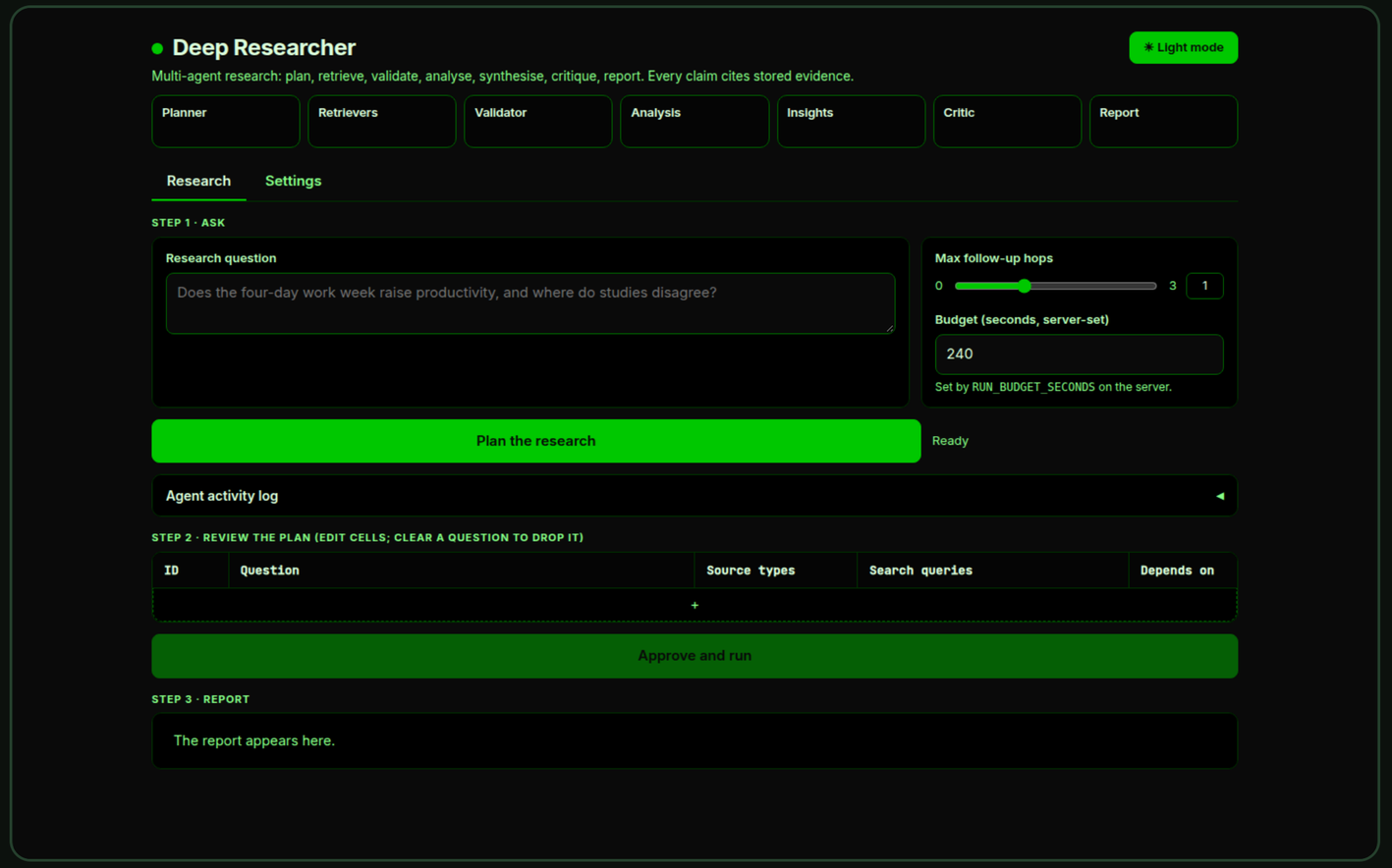
Analysis sends its most valuable gaps back to the retrievers, up to the max-hops limit.

LOOP · RED TEAM

The critic returns insights for revision, or asks for more evidence (capped).

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Three steps from question to report



STEP 1 · ASK

Type the question

Set follow-up hops (0 to 3) and a time budget, then plan.

STEP 2 · REVIEW

Edit the plan

Sub-questions, sources and queries are editable before any retrieval runs.

STEP 3 · REPORT

Read and download

Rendered report with PDF, Markdown and JSON downloads.

The seven pipeline boxes and the activity log update live as each agent works.

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Per-run control, keys stay on the server

A model per tier

Heavy, worker and cheap tiers, picked from the live OpenRouter catalogue with prices.

Reasoning and fallbacks

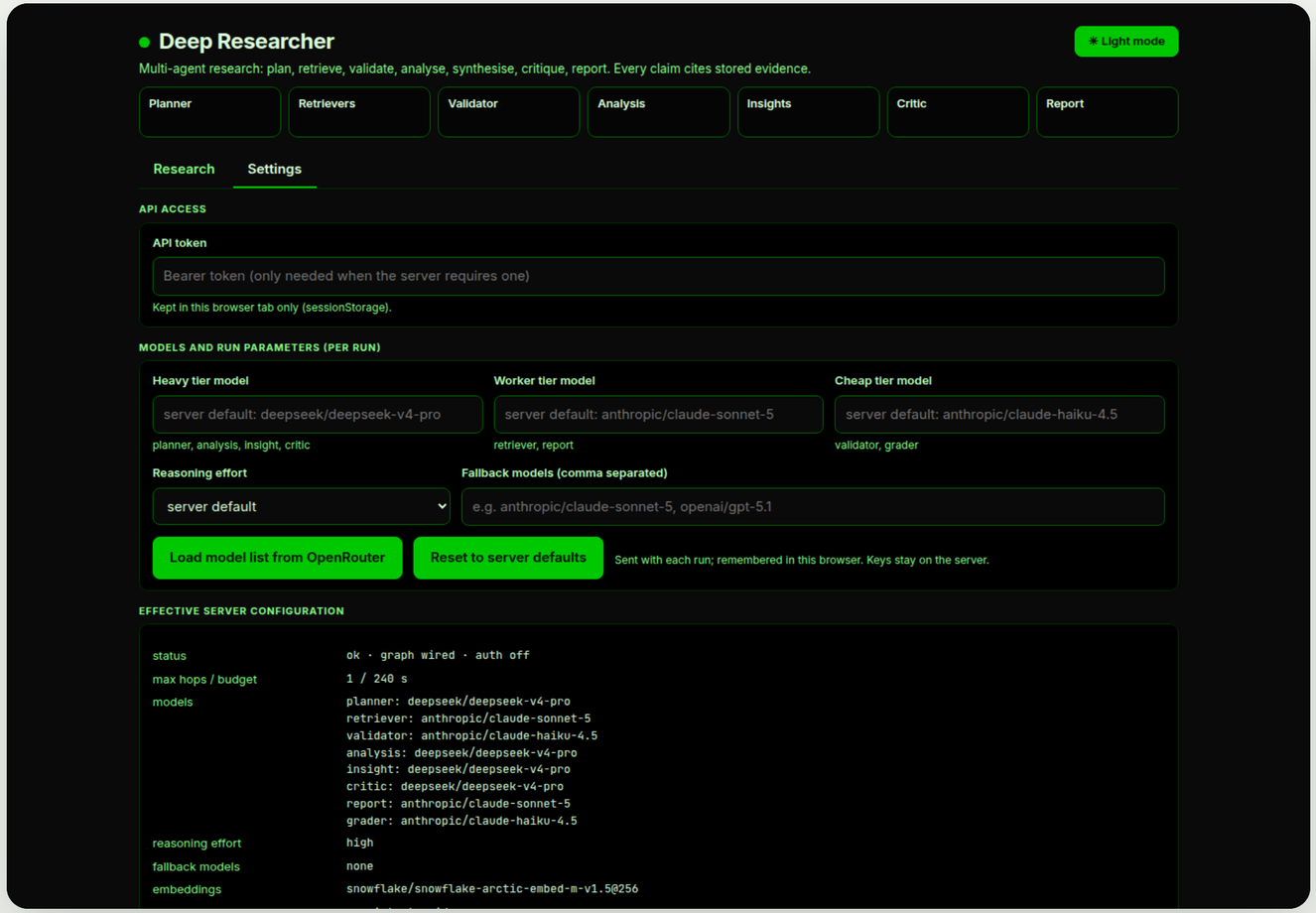
Effort from none to high; backup models tried in order if one fails.

Bounded cost

Budget capped by the server; an optional model allowlist; bearer auth and rate limits.

Nothing hidden

The effective server configuration is shown, and overrides are recorded in the report.



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Every claim traces back to evidence

FROM A GENERATED REPORT

As renewable penetration rises above roughly 50 to 70%, studies indicate a need for storage durations of 8 to 100+ hours [E-776aa3ff21] [E-52ed022ff6]. A policy analysis concludes lithium-ion batteries are viable for daily balancing but not for seasonal storage [E-3206e5befb].

Grid-batteries eval case: 55 sources, 57 claims, 4 contradictions, 5 insights.

Citations are checked

Each evidence ID must resolve to a kept, stored source before the report ships.

Disagreement is surfaced

Contradictions get their own section, and a red-team critic challenges every insight.

Failures are reported

Rate limits, timeouts and model errors land in a Run health section, not a silent log.

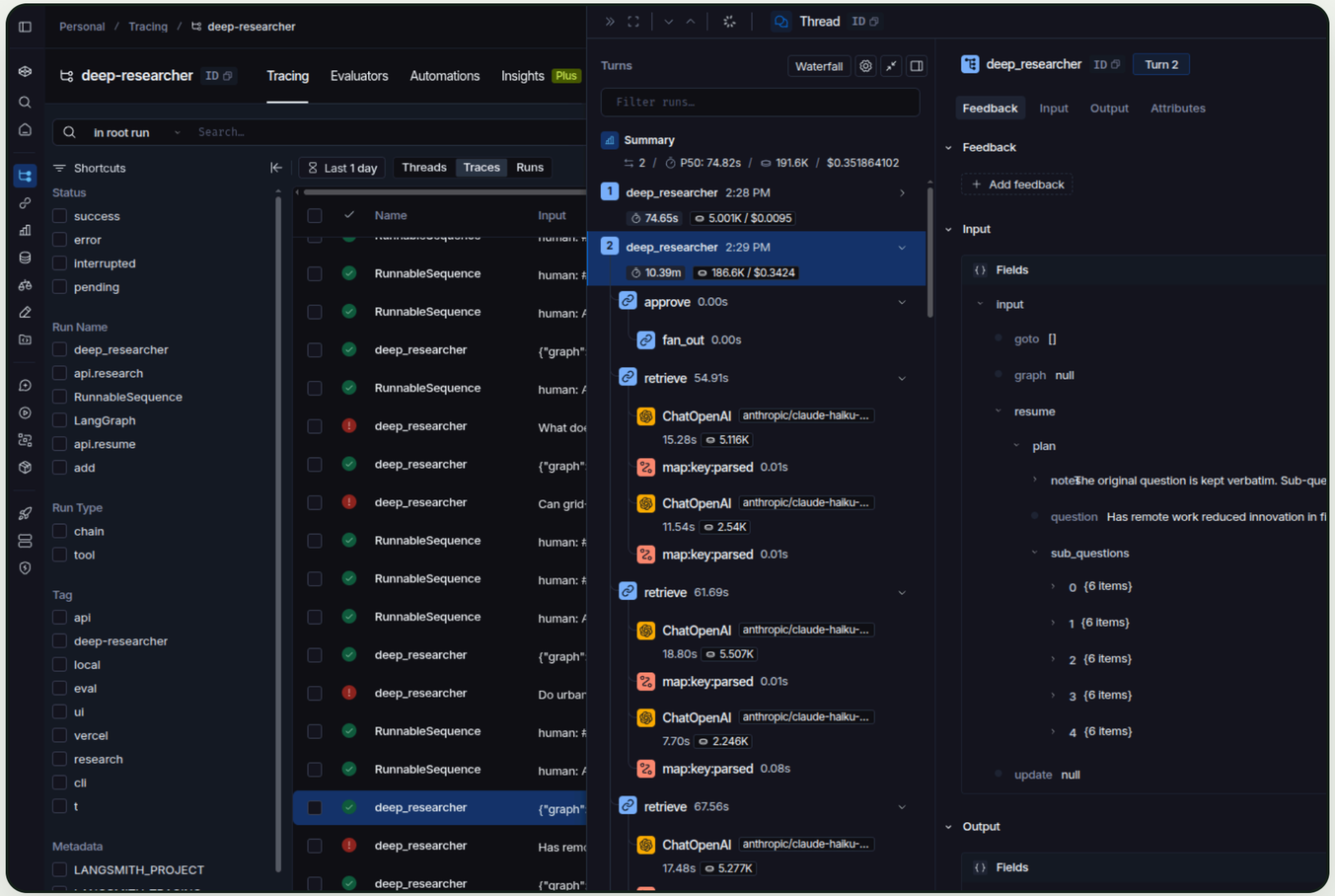
Questions are screened

Guardrails reject profanity and questions that bundle unrelated topics.

SOURCES arXiv | Semantic Scholar | Tavily web and reports | newsdata.io / GDELT | Wikipedia

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Every run is traced in LangSmith



One trace per request

Plan and resume calls are tagged and grouped by thread, so a whole investigation filters into one view.

Every agent nested

Approve, fan-out, parallel retrievers, validator: each with its model, latency and tokens.

Cost at a glance

This research turn: 10.4 min, 186.6K tokens, $0.34.

Linked from the app

The stream's first event carries the trace URL; a no-op when tracing is off.

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EVALUATION · 10 CONTESTED QUESTIONS · BASELINE 2026-10-11

Graded by a judge from a different vendor

100%

Citation integrity

Every cited ID is a kept source

93%

Groundedness

Claims supported by cited text

92%

Coverage

Expected facets addressed

94%

Contradiction recall

Seeded disagreements surfaced

$0.28

model cost per run

12.7 min

mean wall time

55

sources kept per run

Ships anywhere

Local CLI and Gradio, Docker, AWS Lambda, live on Vercel

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